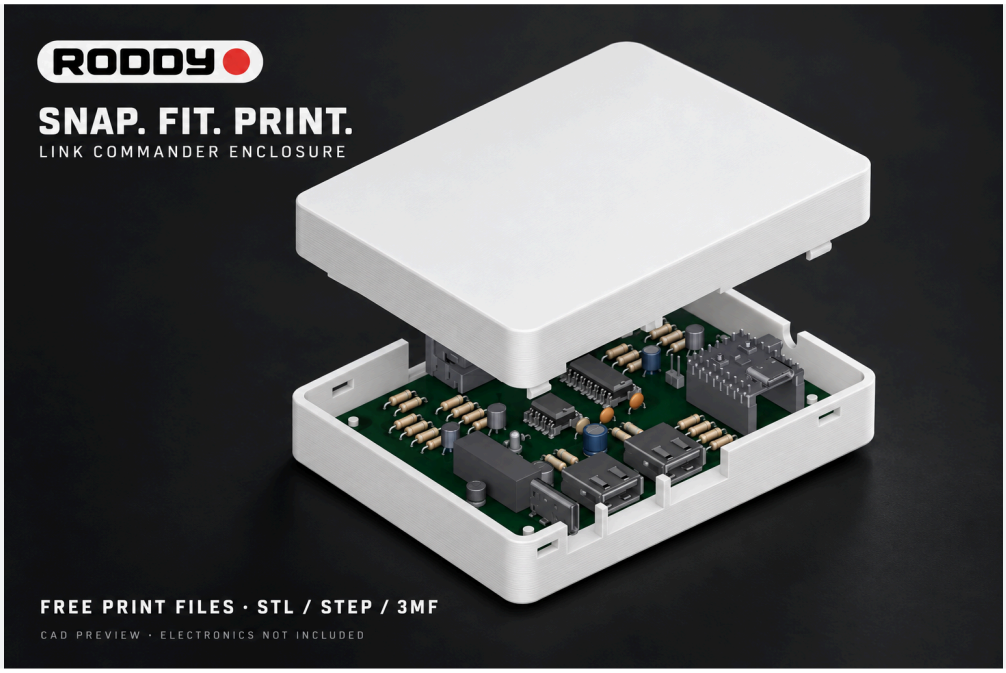
CUSTOMER GUIDE

# PRINT YOUR LINK COMMANDER SHELL




## One shell. Two board revisions.

Free print files for the current 100 × 75 mm Rev E and Rev F boards. The white finish is an example; choose your own unfilled ABS color. No circuit board, electronics, cables or printed parts are included.

## Start small

Print and test the two snap-fit coupon pieces before the full shell. Nominal CAD fit checks passed, but the printed shell, actual components and cable boots still need physical validation.

02 / LINK COMMANDER ENCLOSURE

# Choose your files

## What is in the ZIP

print/BASE_print.stl and LID_print.stl: base and lid oriented for printing. Matching STEP files are also supplied.

print/COMMON_SHELL_X1C_layout.3mf: both shell pieces arranged on a plate. print/SNAP_TEST_layout.3mf: the small mating test pieces. The coupon pieces are also available as individual STL files.

cad/: neutral STEP files for the assembled shell, base and lid. Dimensions are in millimeters. The included PDF is this guide.

## Printer and material

Designed around an enclosed Bambu X1 Carbon with a 0.4 mm nozzle and unfilled ABS. The 3MF files contain model geometry and placement, not ready-to-run G-code or calibrated machine settings. Select your actual printer, filament and build plate in Bambu Studio.

## Orientation

Base: floor down. Lid: broad exterior face against the print plate. Preserve these orientations to give the show face the plate finish and keep support scars off the visible surfaces.

Check the X1C plate and toolhead exclusion areas in your installed slicer. Inspect every layer around the snap fingers, board supports and port windows.

## Fit coupon

Let both pieces cool completely, then engage and release the latch with a plastic pick. Check at least 10 cycles for cracking, whitening or looseness. Do not force a tight snap. Adjust the fit before printing the full enclosure.

03 / LINK COMMANDER ENCLOSURE

# Print settings & finish

## Starting settings, then calibrate

0.4 mm nozzle; 0.16 mm layers; 0.20 mm initial layer. Four walls; eight top and bottom layers; 15% gyroid infill.

Start around 35 mm/s outer-wall speed. Place the seam at a rear corner, away from ports and snap interfaces. Start with an 8 mm outer brim and 0.10 mm brim gap; adjust for your plate.

Elephant-foot compensation starts at 0.15 mm. Avoid aggressive compensation or global scaling: they alter board fit and connector spacing.

## ABS profile

Use the filament maker’s profile for your actual ABS and build plate, including temperatures, drying and adhesive guidance. Keep the printer enclosed and allow gradual cooling. Provide suitable ventilation without cold drafts through the chamber.

Use unfilled ABS for the flexing latches. Carbon- or glass-filled ABS is not the specified material. Do not improvise chamber heaters or disable printer protections.

## Reduce visible layer lines

The broad, flat lid exterior prints against a smooth or satin plate. Vertical sides and rounded corners avoid shallow stepped slopes, although normal FDM side lines remain. A consistent slow outer wall and a hidden seam improve the result.

Do not apply fuzzy skin to mating surfaces, latches or openings. No chemical finishing is required; smoothing can change fit and weaken or distort latch details.

## Before starting

Inspect for supports fused into latch clearances. Confirm short bridges and overhangs with the test pieces before accepting a full print.

04 / LINK COMMANDER ENCLOSURE

# Assemble & reopen

## Fit the board

Disconnect all power and cables. Confirm solder tails cannot touch the plastic. Place either Rev E or Rev F component-side up on the four locating pegs. Check that the board seats freely.

Align two USB-A sockets and power USB-C with the rear openings, two PS/2 sockets with the front windows, and the green LED with the side aperture.

## Close the shell

Lower the lid straight and press near each latch until all four engage. The seam rests on the case walls, not on electronic components. Stop if the lid contacts the module, capacitors, fuse or socketed chips.

The lid now has an integral guard in front of the module USB socket. Check clearance against the real module; it must not touch or press on the connector. The rear 5V POWER label identifies the customer power inlet.

## Open for service

Disconnect power. Use a narrow plastic pick through each latch-release window to press the hook inward while gently lifting that corner. Do not pry on the connector shells or use a metal tool near powered electronics.

JP1 and the module USB are for authorized service only. Disconnect every cable and supply, remove the lid, then remove JP1 before connecting programming USB. Disconnect programming USB before refitting JP1. The plastic guard does not provide electrical isolation.

## Final fit checks

Insert your actual USB and both PS/2 cables. They must fully seat without loading the solder joints or forcing the case. Confirm LED visibility, underside clearance and free access to the connectors.

Check temperatures with the final firmware and intended load before regular use. No thermal qualification or ingress rating is claimed. Keep the enclosure open for early debugging if needed.

05 / LINK COMMANDER ENCLOSURE

# Dimensions & adjustments

## Nominal geometry

Outside: 113.6 × 88.6 × 30 mm. PCB: 100 × 75 × 1.6 mm, with 90 × 65 mm mounting-hole pitch. Walls, floor and roof: 2.4 mm.

PCB underside sits 5 mm above the floor. Locating pegs are 2.6 mm for 3.2 mm board holes. Lid guide clearance is nominally 0.40 mm per side.

## Port openings

PS/2 windows: 18.6 mm square, 20 mm center-to-center. USB-A windows: 18 mm wide. Cable boots vary; dimensions do not prove every plug will fit.

## Adjust locally

Use the STEP geometry for changes. Avoid scaling the entire shell: it changes the distance between the board supports and ports. Change mating clearances and print the test coupon again.

Snap fingers are printed along Z in the prescribed lid orientation. Printed layer bonding and ABS shrinkage affect strength, so repeat the coupon cycle test after changes.

## Help

For updated files, return to the enclosure listing on roddy.world. Email support@roddy.world with your board revision, printer, nozzle, filament and fit observations. This guide describes a prototype design, not a production-qualified enclosure.

For the planned assembled product, the module is installed and programmed and JP1 fitted before the enclosure is closed. Customers use the rear power socket and do not adjust JP1 or handle the module. Firmware and physical qualification remain release requirements.